

HAOYU WANG

✉ haoyu.w.2001@gmail.com · ☎ (+86) 180-3519-9839 · 📄 Google Scholar · 🏠 Homepage

🎓 EDUCATION

Beijing Jiaotong University (BJTU), Beijing, China Sept. 2023 – Jun. 2026

M.Eng. in Electronic Information, School of Computer Science & Technology.

Thesis: X-ray Prohibited Items Detection in Complex Scenes (*Outstanding Master's Thesis*).

Guilin University of Electronic Technology (GUET), Guilin, China Sept. 2019 – Jun. 2023

B.Eng. in Robotics Engineering, School of Artificial Intelligence.

Thesis: An Object-Detection-Based Assistant for the Visually Impaired (*Outstanding Bachelor's Thesis*).

💡 RESEARCH INTERESTS

Neuro-symbolic AI · Controllable and interpretable AI · Continual learning.

📄 PUBLICATIONS

Accepted

- R. Tao, **H. Wang**, Y. Guo, H. Chen, L. Zhang, X. Liu, Y. Wei, Y. Zhao. “Dual-view X-ray Detection: Can AI Detect Prohibited Items from Dual-view X-ray Images like Humans?” *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- Z. Li, **H. Wang**, W. Wang, C. Tan, Y. Wei, Y. Zhao. “DCI: Dual-Conditional Inversion for Boosting Diffusion-Based Image Editing.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Z. Liu, J. Zhang, **H. Wang**, B. Xu, C. Zhang, Y. Zhu, E. Shi. “Scaling Law for Large Wireless Models.” *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.

Under Review

- **H. Wang**[†], R. Tao[†], W. Wang, Y. Wei, Y. Zhao. “PAD-F: Prior-Aware Debiasing Framework for Long-Tailed X-ray Prohibited Item Detection.” *IEEE Transactions on Information Forensics and Security (IEEE TIFS)*, major revision. ([†] *Equal contribution*)

🏢 RESEARCH EXPERIENCE

X-ray Prohibited Object Detection

Sept. 2023 – Jun. 2026

- **Dual-view detection with auxiliary-view enhancement (CVPR 2025).** Designed a dual-branch detection pipeline inspired by how human inspectors consult the side view to disambiguate occluded items: the side view is processed as an auxiliary stream that specifically targets challenging instances missed by the main view, and its predictions are fused back to complement main-view detections. Ran all experiments; co-built and released the first large-scale dual-view X-ray prohibited-item dataset.
- **Prior-aware debiasing for long-tailed X-ray detection (IEEE TIFS, major revision).** Proposed the debiasing framework, ran all experiments, and co-wrote the paper. For augmentation, leveraged X-ray physical-imaging priors to match material-consistent foreground instances with plausible host regions, synthesising challenging yet physically realistic training samples for tail categories. For classification, exploited implicit category co-occurrence priors mined from high-confidence proposals to improve detection performance.

Concept learning and structured benchmarks

Nov. 2025 – Present

- **Probing concept-level compositionality of pretrained visual representations.** On top of frozen DINOv3 features, trained sparse autoencoders to decompose the continuous representation into discrete concept units, and used rule heads to explicitly compose these concepts into per-class decision rules, testing to what extent such representations can actually be used through explicit concept composition. Leading the planning and experiments of this ongoing study.

- **A 3D visual benchmark with controllable compositional structure.** Building a synthetic benchmark driven by an explicit grammar: every object is generated by composing modular parts, attributes, and spatial relations according to grammar rules. Built an automated generation-and-annotation pipeline on top of AI-generated assets, as a testbed for systematically evaluating the compositional generalisation of representation-learning and structured methods. Driving the project from benchmark design through experiments, with development ongoing. Project page: [Bestiary-Engine](#).

LLMs, symbolic learners and agents

May 2026 – Present

- **Relieving hypothesis-space explosion in ILP with LLM-proposed hypotheses.** Asked whether a language model can propose high-quality hypotheses and thereby relieve the hypothesis-space explosion in inductive logic programming. In a prototype study the model was given exactly the input the Popper solver receives — background knowledge, examples and language bias, with the permitted number of variables deliberately relaxed — and its proposals were compared against Popper’s solving time. On each of the three tasks tried, a hypothesis consistent with B and E appeared sooner than Popper found one, and on some tasks it was also smaller ($1608.6\text{ s} \rightarrow 216.4\text{ s}$ on `synthesis-filter`; $350.3\text{ s} \rightarrow 193.4\text{ s}$ with size 42 $\rightarrow 7$ on `synthesis-finddup1`) — a positive signal for bringing language models into the ILP loop, with the caveat that the study used one model, one seed and no held-out test set. Designed and ran the study.
- **Video games as a benchmark for LLM agents.** Studying released commercial games as a benchmark for LLM agents: comparing a GUI agent driven by a live model against an arm in which the agent’s knowledge is frozen into executable operating code, in order to see how far code written by an agent can go in actually playing these games. Main contributor to the problem formulation and the development of the codebase.

🔧 PROJECTS

SurMount: Wearable smart-camera assistant for the visually impaired May 2021 – May 2023

Guangxi Provincial Undergraduate Innovation and Entrepreneurship Program. Designed a wearable system (vest-mounted compute + headband camera) running a lightweight YOLO detector on-device. Proposed an annotation-free, per-user personalization pipeline: a short multi-view video of each target object, captured at enrollment, is automatically decomposed into multi-view training crops and used to fine-tune a compact, object-specific detector, selected at inference via voice command.

Continual Reinforcement Learning with Neuro-Symbolic Components Apr. 2025 – Sept. 2025

Built a world-model continual-RL agent (following the Continual Dreamer framework) on the MiniHack benchmark, trained and evaluated on a continual task sequence. On top of this agent, experimented with using neuro-symbolic methods such as a *probabilistic logic shield* and a *symbolic Markov transition prior (NeSy-MM)* to inject prior knowledge into the continual RL pipeline, under varied constraint and shield configurations.

🏆 HONOURS & AWARDS

First-Class Academic Scholarship , BJTU (<i>three consecutive years</i>)	2023 – 2026
Outstanding Master’s Thesis , BJTU	2026
Outstanding Bachelor’s Thesis , School of Artificial Intelligence, GUET	2023
Outstanding Completion (highest rating), Provincial Undergraduate Innovation Program	2023

👥 ACADEMIC SERVICE

Conference reviewer: AAAI, ICME, ICPR.

Journal reviewer: IEEE TMM.

🔗 TECHNICAL SKILLS

Python · PyTorch · Linux · Git · $\LaTeX$.